

Notes: Ma (RFS, 2025)

Technological Obsolescence

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1 Big picture

- **Question.** How can we directly measure the *destruction* side of creative destruction — the fading relevance of a firm’s existing technologies — and what does that measure tell us about firm performance and stock returns?
- **Core contribution.** The paper builds a new firm-year patent-based measure of *technological obsolescence* using changes in citations to a firm’s *technological base* rather than the arrival of new patents. The measure is designed to track how far a firm’s existing knowledge stock moves away from the technology frontier.
- **Why this is conceptually new.** Most innovation measures are flow measures of *new* innovation. This paper studies the stock side: what happens to the much larger set of ideas a firm already accumulated. The paper emphasizes that annual new patents are only a small share of the existing patent portfolio, so the evolution of old knowledge is economically important.
- **Main real-side finding.** Higher technological obsolescence predicts lower future profits, output, capital growth, employment growth, and revenue-based productivity. The effect is strongest when obsolescence hits *core* technologies and *product* / *embodied* innovations, and it is larger in more competitive product markets.
- **Main asset-pricing finding.** High-obsolescence firms underperform low-obsolescence firms in the stock market. A strategy that buys low-obsolescence firms and shorts high-obsolescence firms earns large and statistically significant alphas across standard factor models.
- **Mechanism for returns.** The paper argues that markets underreact to technological obsolescence. Analysts systematically overestimate the future earnings of firms whose technologies are becoming obsolete.
- **Why the paper matters.** The paper gives the literature a direct and scalable measure of innovation depreciation at the firm-year level. It also links innovation dynamics to both real reallocation and financial mispricing.

2 Data and measurement (patents, citations, and the obsolescence variable)

2.1 Patent data and sample construction

- **Patent data.** The paper uses USPTO patent data from 1976 to 2020, covering nearly seven million patents.
- **Link to firms.** Patents are linked to Compustat public firms using the NBER bridge, the extended KPSS data, and fuzzy matching for later years.
- **Main firm sample.** The core empirical analysis focuses on U.S. public firms from 1986 to 2016. Starting in 1986 gives enough pre-period patent history to construct the measure; stopping in 2016 mitigates right-truncation in future citations.
- **Information observed.** For each patent, the paper observes backward citations (what prior patents the patent builds on) and forward citations (what later patents cite it and when).

- **Innovation-focused sample.** Many analyses focus on firms that are meaningfully innovative — firms that at some point receive at least 10 patents — though the paper notes that the core conclusions are robust more broadly.

2.2 Step 1: The technological base

- **Definition.** A firm’s technological base at time $t - \omega$ is the set of all patents cited by the firm’s own innovation up to $t - \omega$, excluding patents owned by the focal firm.
- **Economic meaning.** This base proxies for the stock of underlying technologies and knowledge the firm has been standing on. The paper compares it to a researcher’s knowledge base represented by the papers and books cited in prior research.
- **Why exclude own patents?** Excluding the firm’s own patents makes the measure more arms-length from firm-specific operational shocks and from the firm’s own current innovation choices.
- **Size and persistence.** The technological base is large and persistent. The paper reports an average of about 2,001 patents in the base (median 219), and the base expands only slowly over time.
- **Within-industry heterogeneity.** Even firms in the same SIC3 industry often rely on very different technological bases. More than 90% of within-industry firm pairs have zero overlap in their technological bases.

2.3 Step 2: Citation dynamics around the base

- **External citations only.** For a fixed technological base, the paper counts citations the base receives in year $t - \omega$ and in year t , excluding citations made by the focal firm itself.
- **Interpretation.** If a patent in the base continues to receive citations from later patents, it remains useful for new innovation. If it stops being cited, the technology is becoming obsolete.
- **Why citations are informative.** Figure 2 shows that new patents most often cite patents roughly 4 to 10 years old, and citations decay for older vintages. This age pattern is consistent with technological aging and obsolescence.

2.4 Step 3: The obsolescence measure

- **Core idea.** Compare how much the same technological base is cited today versus ω years earlier.
- **Sign convention.** A *positive* value of obsolescence means the base now receives fewer citations than before; that is, the firm’s inherited knowledge stock has become less relevant.
- **Example from the paper.** If the base received 1,000 external citations in 2003 and 900 in 2005, the obsolescence measure equals about +10%. If citations instead rose to 1,100, obsolescence would be −10%, indicating the firm’s base moved closer to the frontier.
- **Share-shift flavor.** The measure combines a predetermined firm-specific exposure ($\mathcal{B}_{f,t-\omega}$) with external technological evolution ($\text{Cit}_t - \text{Cit}_{t-\omega}$). That is exactly why the author describes it as share-shift-like.

2.5 Descriptive properties of the measure

- **Average magnitude.** The mean obsolescence is 7.843% for $\omega = 1$, 12.900% for $\omega = 3$, 19.390% for $\omega = 5$, and 34.126% for $\omega = 10$.
- **Wide dispersion.** For the one-year measure, the 10th percentile is -8.039% and the 90th percentile is 24.202%. So some firms' technological bases become *more* relevant while others experience sharp disruption.
- **Within-firm and within-industry variation.** Around 60% of total variation is within-firm over time, and a large share of the remaining variation is across firms within the same industry-year.
- **Average annual depreciation rate.** The paper summarizes the economic magnitude as roughly 4%–7% annual obsolescence, broadly consistent with aggregate estimates of patent-value depreciation.

2.6 What drives technological obsolescence?

- **Three candidate sources.** The paper studies whether obsolescence comes from the firm's own new innovation, competitors' innovation, or broader innovation outside the firm and even outside the industry.
- **Table 2 result.** All three sources matter. Firm-own innovation is strongly positively associated with obsolescence, consistent with cannibalization. Competitors' innovation and economy-wide / upstream innovation also help explain obsolescence.
- **Interpretation.** The measure is not just “my rivals innovated.” It captures multiple routes through which a firm's existing technologies can lose relevance.

2.7 Comparison with measures of new innovation

- **Measures compared.** The paper compares obsolescence with KPSS patent value (SM), citation-weighted patents (CW), Kelly et al. breakthrough patents, Bowen-Frésard-Hoberg's RETECH and tech breadth measures, and Bellstam-Bhagat-Cookson's text-based innovation.
- **Correlation pattern.** Obsolescence is negatively but only mildly correlated with these new-innovation measures: for example, correlations are -0.061 with SM, -0.085 with CW, -0.148 with breakthrough patents, and -0.083 with text-based innovation.
- **Why this matters.** The weak negative correlations show that the paper is not simply constructing another “low innovation” proxy. It is measuring something related but distinct: deterioration of existing technology.
- **Contrast with new-innovation measures.** The new-innovation measures are positively correlated with one another, usually more strongly than with obsolescence.

2.8 Validation exercises

- **Figure 2: citation age profile.** Backward citation lags peak around 4 to 10 years, then decay. This validates citations as a sensible way to track the life cycle of technologies.

- **Figure 3: HDD case study.** In the hard-disk-drive transition from 5.25-inch to 3.5-inch technology, patents tied to old 5.25-inch designs show rising obsolescence, while patents tied to 3.5-inch designs do not. This is a clean industry example where the direction of technological replacement is well known.
- **Figure 4: breakthrough innovation.** When a firm’s technology class experiences a breakthrough innovation as identified by Kelly et al. (2021), the firm’s obsolescence jumps after the breakthrough. The pretrend is flat.

2.9 Alternative constructions and robustness of measurement

- **Obsolescence from declining patents only.** The author reconstructs the measure using only patents in the technological base that actually experienced citation declines. The main results remain strong.
- **Trend adjustment.** The paper adjusts for the secular rise in patenting and citations over time. Results are robust.
- **Patent aging / expiration.** The paper either excludes expired patents or discounts citations to old patents. Results are again robust.
- **Alternative own-patent measure.** A natural alternative is to track citation declines to the firm’s own patents rather than to its technological base. That alternative is positively correlated with the main measure (correlation 0.361) and often produces even stronger results, but it is noisier and more endogenous.

3 Models and empirical specifications (all equations + interpretation)

3.1 Technology base and citation counts

Let $\mathcal{P}_{f,t-\omega}$ denote the firm’s own patent portfolio accumulated by $t - \omega$. The paper’s technological base can be written schematically as

$$\mathcal{B}_{f,t-\omega} = \{p : p \text{ is cited by some patent of firm } f \text{ by } t - \omega, p \notin \mathcal{P}_{f,t-\omega}\}. \quad (1)$$

For a fixed base, define the external citation count in year s as

$$\text{Cit}_s(\mathcal{B}_{f,t-\omega}) = \sum_{p \in \mathcal{B}_{f,t-\omega}} \text{external forward citations received by patent } p \text{ in year } s. \quad (2)$$

- **Interpretation.** $\mathcal{B}_{f,t-\omega}$ is the predetermined stock of technologies the firm had been drawing on. $\text{Cit}_s(\mathcal{B}_{f,t-\omega})$ measures how relevant that stock still is for later innovation.
- **Why this construction is clever.** The firm’s technological exposure is fixed at $t - \omega$, so changes in citations from $t - \omega$ to t mostly reflect external technological evolution rather than contemporaneous choices by the focal firm.

3.2 Main measure of technological obsolescence

The central variable in the paper is

$$\text{Obsolescence}_{f,t}^{\omega} = -\left[\ln \text{Cit}_t(\mathcal{B}_{f,t-\omega}) - \ln \text{Cit}_{t-\omega}(\mathcal{B}_{f,t-\omega})\right]. \quad (3)$$

- **Interpretation.** This is the ω -year log decline in citations to the same technological base. A larger positive value means the firm's inherited technologies are losing relevance faster.
- **Why the log form helps.** The measure is a within-firm growth rate, so it naturally differences out firm size and the absolute size of the knowledge space.
- **Economic meaning.** A firm is doing badly in the obsolescence sense not because it currently patents less, but because the technologies it had previously relied on are being left behind.

3.3 Baseline real-outcome specification

The main firm-growth regression is

$$\log Y_{f,t+\tau} - \log Y_{f,t} = \beta_{\tau} \text{Obsolescence}_{f,t} + \theta_{\tau} X_{f,t} + \delta_{I \times t} + \varepsilon_{f,t+\tau}. \quad (4)$$

- **Dependent variables.** The paper alternates among profits, output, capital, employment, and revenue-based TFP.
- **Regressor of interest.** $\text{Obsolescence}_{f,t}$, usually measured over the previous five years ($\omega = 5$), standardized to unit standard deviation.
- **Controls.** The paper controls for the current level of the outcome, log capital, log employment, log patent stock, firm age, and idiosyncratic volatility.
- **Fixed effects.** SIC3-by-year fixed effects, so identification is from firms within the same narrow industry and year.
- **Inference.** Standard errors are clustered by firm and year.
- **Interpretation of β_{τ} .** A negative β_{τ} means that firms with more rapidly obsolescing technologies today grow more slowly in the future.

3.4 Augmented growth specifications: new innovation and competitor innovation

The paper augments the baseline specification by adding measures of new innovation and competitor innovation:

$$\log Y_{f,t+\tau} - \log Y_{f,t} = \beta_{\tau} \text{Obsolescence}_{f,t} + \gamma_{\tau} \text{SM}_{f,t} + \kappa_{\tau} \text{CW}_{f,t} + \theta_{\tau} X_{f,t} + \delta_{I \times t} + \varepsilon_{f,t+\tau}, \quad (5)$$

and in another comparison,

$$\log Y_{f,t+\tau} - \log Y_{f,t} = \beta_{\tau} \text{Obsolescence}_{f,t} + \gamma_{\tau} \text{SM}_{f,t} + \rho_{\tau} \text{SM}_{f,t}^{\text{competitor}} + \theta_{\tau} X_{f,t} + \delta_{I \times t} + \varepsilon_{f,t+\tau}. \quad (6)$$

- **Purpose.** These specifications test whether technological obsolescence contributes information beyond standard measures of new innovation and beyond leave-me-out competitor innovation.

- **Reading of coefficients.** If β_τ remains stable after adding SM, CW, and competitor SM, then obsolescence is not just a relabeling of these earlier measures.

3.5 Portfolio sorts and factor-model alphas

For the asset-pricing part, firms are sorted each June into Low, Middle, and High obsolescence portfolios based on prior-calendar-year obsolescence. The paper then estimates standard factor-model regressions of portfolio excess returns:

$$R_{p,m}^e = \alpha_p + \beta_p' F_m + u_{p,m}, \quad (7)$$

where F_m is alternatively the Fama-French three-factor model, four-factor model, five-factor-plus-momentum model, the Hou-Xue-Zhang q-factor model, or models that replace HML with intangible-adjusted HML.

- **Portfolio construction.** Low means below the 30th percentile of obsolescence, High means above the 70th percentile, and portfolios are held for the next 12 months.
- **Key object.** The paper focuses on the long-short spread Low – High.
- **Interpretation of α_p .** A positive alpha for Low-Minus-High means high-obsolescence stocks are too expensive today relative to future cash-flow news and therefore underperform later.

3.6 Analyst-forecast-error specification

To test the underreaction mechanism, the paper estimates

$$\frac{\pi_{f,t+\tau} - F_t \pi_{f,t+\tau}}{P_{f,t-1}} = a + b_{t+\tau} \text{Obsolescence}_{f,t-1} + \varepsilon_{f,t+\tau}, \quad \tau \in \{1, 2\}. \quad (8)$$

- **Dependent variable.** Realized EPS minus the consensus analyst EPS forecast, scaled by lagged stock price.
- **Forecast timing.** The paper uses forecasts issued within 3 months after annual earnings announcements and forms the firm-level consensus using the median analyst forecast.
- **Interpretation of $b_{t+\tau}$.** If $b_{t+\tau} < 0$, analysts are too optimistic about firms with high obsolescence. That is direct evidence in favor of underreaction and mispricing.

3.7 Alternative obsolescence construction

A natural alternative is to track annual citations to the firm's own patents:

$$\widetilde{\text{Obsolescence}}_{f,t}^\omega = -\left[\ln \text{Cit}_t(\mathcal{P}_{f,t-\omega}) - \ln \text{Cit}_{t-\omega}(\mathcal{P}_{f,t-\omega})\right]. \quad (9)$$

- **Why it is attractive.** This directly tracks the changing relevance of the firm's own patent portfolio.
- **Why the paper does not make it the main measure.** The firm's own patent portfolio is much smaller and therefore citation dynamics are bumpier. More importantly, citation changes

to the firm's own patents are more exposed to contemporaneous firm-specific shocks and therefore more endogenous for firm-level regressions.

4 Identification and instruments (what makes the empirical interpretation credible)

4.1 This is not an IV paper

- The paper does **not** claim a fully causal reduced-form estimate of the effect of obsolescence on firm outcomes.
- There is no external instrument and no quasi-experimental identification in the usual sense.
- The correct way to read the paper is as a very careful measurement paper with strong validation and predictive regressions.

4.2 Why the obsolescence measure is plausibly informative

- **Predetermined exposure.** The technological base is fixed at $t - \omega$, before future firm outcomes are measured.
- **External citation dynamics.** The paper excludes the firm's own patents from the base and excludes self-citations when measuring citation evolution.
- **Within-industry-year design.** Industry-year fixed effects absorb common technology and demand conditions, so the remaining variation compares firms in the same product market but with different inherited technology exposures.
- **Substantial cross-firm heterogeneity in exposure.** Because firms' technological bases overlap little even within the same SIC3 industry, the within-industry-year comparisons are economically meaningful rather than mechanically trivial.

4.3 Why this is not just low new innovation

- Obsolescence is only mildly negatively correlated with the leading measures of new innovation.
- When the paper controls for SM and CW, the coefficient on obsolescence barely moves.
- This is an important credibility point: obsolescence is measuring deterioration of *past* technologies, not simply a failure to generate new patents today.

4.4 Validation evidence

- **Figure 2.** The citation-age profile shows the raw life-cycle pattern the paper is exploiting.
- **Figure 3.** In the HDD transition, the measure moves exactly where industry history says technological replacement occurred.
- **Figure 4.** Around external breakthrough innovations, firms exposed to the disrupted technology experience a discrete rise in obsolescence.

4.5 Robustness against endogeneity

- **Only declining-base patents.** The paper reconstructs the measure using only the subset of patents that actually lose citations. Results survive and often strengthen.
- **Patent-age adjustments.** Results are robust to excluding expired patents and discounting older patents.
- **Trend adjustment.** Results are robust to correcting for the secular rise in patenting intensity and hence the secular rise in citations.
- **Firm fixed effects.** The results remain when firm fixed effects are added, so they are not purely cross-sectional.
- **Excluding market competitors.** The technological base and citation dynamics can be reconstructed excluding product-market competitors; the results are robust.
- **Scientific / arms-length bases.** The author further strengthens the interpretation by building the base using only more scientific and less firm-specific patents: general-purpose technologies, international patents, university and government patents, and standard-essential patents.

4.6 How to read the stock-return evidence

- **Main interpretation.** The paper's preferred reading is mispricing via underreaction to a slow-moving and complicated form of technology deterioration.
- **Direct supporting evidence.** Analyst forecast errors are systematically more optimistic for high-obsolescence firms.
- **Alternative preference-based explanation.** The paper examines a prospect-theory-based explanation and concludes that it can explain only about 5%–10% of the observed alpha spread.
- **What remains uncertain.** Even with the analyst evidence, the paper cannot rule out all risk-based stories. So the asset-pricing interpretation is persuasive but not ironclad.

5 Tables and figures

5.1 Figure 1: Existing patent portfolio, obsolescence, and new innovation

Read:

- The figure is conceptual, but it is extremely useful. It separates three objects that are often conflated: the stock of existing patents, the fading of those patents' relevance, and the arrival of new patents.
- The paper's whole point is that the fading of the existing stock should be measured separately from new innovation.
- This figure is the cleanest visual summary of the paper's conceptual contribution.

5.2 Figure 2: Patent backward citation dynamics

Read:

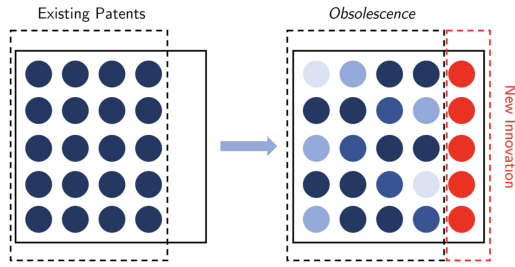


Figure 1
Illustration of existing patent portfolio, obsolescence, and innovation
This figure illustrates the differences between the existing patent portfolio, its obsolescence, and the arrival of new innovation.

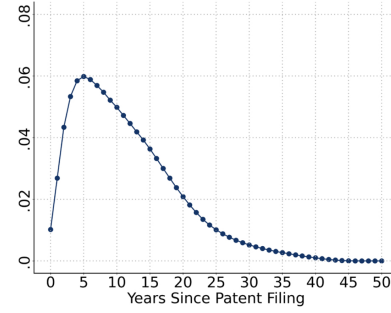


Figure 2
Patent backward citation dynamics
This figure plots distributions of backward citation lags. Specifically, each data point in the data is a citation pair: the citing patents and the cited. It plots the distribution of the age of the cited patents at the time for which the citing patent was applied.

- New patents are most likely to cite patents that are around 4 to 10 years old.
- Citation intensity then declines with age, which is exactly the raw pattern the obsolescence measure needs in order to be meaningful.
- Economically, the figure says that even successful ideas have a technological life cycle.

5.3 Figure 3: Obsolescence of example HDD patents

Read:

- The figure compares patents linked to old 5.25-inch HDD technology with patents linked to newer 3.5-inch technology.
- During the industry transition in the late 1980s and early 1990s, the 5.25-inch patents show sharply rising obsolescence, while the 3.5-inch patents are much flatter.
- This is a strong face-validity check because the direction of technological replacement in HDDs is well documented outside the paper.

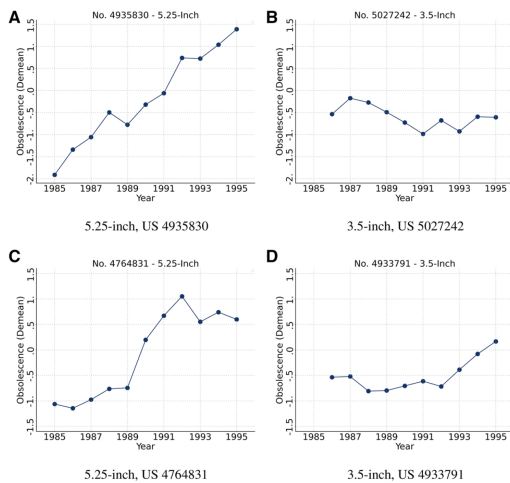


Figure 3
Obsolescence of example HDD patents
This figure plots the obsolescence measure for example HDD patents. Patent numbers and relevant HDD generations (5.25-inch and 3.5-inch) are provided in the subfigures.

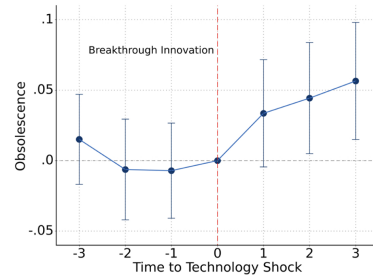


Figure 4
Obsolescence in response to breakthrough innovation
This figure shows the change of *Technological obsolescence* in firms that experience a breakthrough innovation in technology classes that the firm innovates in. The arrival of breakthrough innovation follows Kelly et al. (2021), who use textual information of patent filings.

5.4 Figure 4: Obsolescence in response to breakthrough innovation

Read:

- Firms exposed to a breakthrough innovation in their technology class experience a jump in obsolescence after the shock.
- The pretrend is approximately flat, which helps the interpretation.
- This figure extends the validation beyond one industry and shows that the measure reacts in the right direction when outside technological shocks arrive.

5.5 Table 1: Summary statistics

Read:

- The one-year obsolescence measure has mean 7.843%, standard deviation 13.384, 10th percentile -8.039%, and 90th percentile 24.202%.
- The five-year measure has mean 19.390%, which implies roughly 3.9% annual obsolescence over a five-year window.
- The distribution is wide, which is exactly what the paper needs in order to study winners and losers from technological change.
- The table also makes clear that the sample firms are patent-intensive and economically large, so the paper is about innovative public firms rather than the average small firm.

Table 1
Firm-year-level summary statistics

	Count	Mean	SD	10%	25%	50%	75%	90%
A. Innovation measures								
Obsolescence, horizon $h = 1$ (%)	32,697	7.843	13.384	-8.039	-0.414	7.438	15.667	24.202
Obsolescence, horizon $h = 3$ (%)	32,697	12.900	26.557	-19.869	-3.336	13.346	29.456	44.768
Obsolescence, horizon $h = 5$ (%)	32,697	19.390	34.965	-22.926	-1.626	19.865	40.465	62.389
Obsolescence, horizon $h = 10$ (%)	30,644	34.126	51.844	-28.776	0.867	32.694	65.797	100.861
Stock market-based patent value (\$M) (%)	32,697	15.322	33.402	0.000	0.251	3.699	15.412	41.504
Citation-weighted patents (CW) (%)	32,697	7.788	20.317	0.000	0.095	1.484	5.926	17.739
Competitors' patent value (%)	32,697	25.022	32.516	0.455	4.803	17.398	30.553	50.144
Competitors' citation-weighted patents (%)	32,697	2.840	2.498	0.075	0.687	1.853	4.594	7.073
B. Firm performance and characteristics								
Profit, growth rate (%)	29,734	4.132	29.619	-24.289	-6.369	4.575	15.529	32.283
Output, growth rate (%)	31,261	3.859	31.495	-23.199	-6.019	4.215	14.769	31.487
Capital, growth rate (%)	31,995	6.173	19.795	-8.381	0.140	5.520	12.548	23.870
Labor, growth rate (%)	31,654	2.021	21.017	-16.705	-5.215	1.660	9.473	22.314
TFP, growth rate (%)	23,816	-0.810	25.809	-24.325	-9.867	-0.415	9.179	22.837
Profit	32,689	2,190,551	5,878,533	7,210	48,198	260,424	1,260,023	5,202,854
Output	32,290	6,285,039	16,637,534	27,621	132,299	723,575	3,766,237	15,724,789
Capital	32,596	4,388,214	15,874,031	11,279	44,204	259,103	1,774,694	8,185,960
Labor	32,342	17,702	41,684	0.133	0.520	2.466	13.500	47.114
TFP	23,055	-0.136	0.448	-0.798	-0.530	-0.332	-0.113	0.166
Patent stock	32,697	399,323	1,998,837	15	25	59	243	1,165
Firm age	32,697	28.374	15.646	11	16	25	38	49
Idiosyncratic volatility	32,623	0.001	0.002	0.000	0.000	0.001	0.001	0.003

This table summarizes firm-year level characteristics. Obsolescence measures are defined in Equation (1), and we report the measures with four different parameters, $h = 1, 3, 5, 10$. Stock market-based patent value is based on Kaplan, Paguilloles, Serva, and Stoffman (2017), capturing the stock market reactions to new patent approval. Citation-weighted patent counts (CW) is the total forward citations received by the patents for which a firm applies, and subsequently receives, in each year. Competitors' SM and CW are based on the least-net-out ratio of the SM and CW measure of firms in the same SIC industry in the same year. We report the growth rate and raw values of the following variables: Profit is firm gross profit (CompuStat item sale minus CompuStat item cogs, deflated by the CPI); Output is the nominal value of firm output (CompuStat item sale plus change in inventories CompuStat item invt, deflated by the CPI); Capital is firm capital stock (CompuStat item pape, deflated by the MIPA price of equipment); Labor is the number of employees (CompuStat item emp); TFP is revenue-based productivity constructed based on the methodology of Olley and Pakes (1996) using the procedure in Imbens and Todd (2014); Patent Stock is the number of patents filed by the firm up to that year (using the right truncation problem); Firm Age is the age of the firm starting from the time that the firm's first patent application or entered the CompuStat; Idiosyncratic Volatility is the realized mean idiosyncratic squared returns, where the firm's idiosyncratic return is defined as the firm's return minus the return on the market portfolio. Detailed variable definitions are provided in the Appendix. Variables are winsorized at 1% and 99% using annual breakpoints.

Table 2
Sources of technological obsolescence

	(1)	(2)	(3)	(4)
	Obsolescence			
<i>Firm's own new patent value</i>	0.052*** (0.011)	0.079*** (0.015)	0.072*** (0.014)	0.104*** (0.020)
<i>Competitors' patent value</i>	0.050*** (0.022)	0.042 (0.025)	-0.004 (0.044)	0.018 (0.035)
<i>Upstream effects of innovation</i>	0.041 (0.026)	0.057** (0.027)		
<i>Economy-wide index of innovation</i>			0.108** (0.052)	0.098** (0.037)
Industry FE	Yes	No	Yes	No
Firm FE	No	Yes	No	Yes
Year FE	Yes	Yes	No	No
Observations	27,671	27,671	28,049	28,049
R ²	0.271	0.505	0.132	0.408

This table shows the correlations between the *Obsolescence* measure and potential sources of new innovation, including a firm's own new innovation (*Patent value*), a firm's industry rivals' new technological breakthroughs (*Competitors' patent value*), and innovation from outside the boundary of the specific industry (*Economy-wide index of innovation* or *Upstream effects of innovation*). The *Firm's own new patent value* and *Competitors' patent value* is calculated using the average value in the past 5 years, and *Economy-wide index of innovation* and *Upstream effects of innovation* is measured 6 years ago. *Economy-wide index of innovation* is calculated following Kogan et al. (2017), and *Upstream effects of innovation* is calculated in an external network following Acemoglu, Akcigit, and Kerr (2016) except that we use patent value instead of patent number. Control variables include the log value of average total assets in the past 5 years, the log value of the average book-to-market ratios in the past 5 years, and the average value of the firm's idiosyncratic volatility in the past 5 years. All right-hand-side variables are standardized to unit standard deviation to facilitate magnitude interpretations. Standard errors are clustered by firm and year, and they are shown in parentheses. *, **, and *** denote statistical significance at the 10%, 5%, and 1% levels, respectively.

5.6 Table 2: Sources of technological obsolescence

Read:

- A firm's own new patent value enters positively and robustly, consistent with within-firm cannibalization of old technology by new technology.
- Competitors' patent value also matters, especially in the industry-year specification.
- Upstream innovation and the economy-wide innovation index are also positively associated with obsolescence, showing that disruption can come from outside the focal industry.

- Takeaway: obsolescence is multi-source. That is one reason the paper’s measure is richer than simple leave-me-out competitor innovation.

5.7 Table 3: Relation between obsolescence and new-innovation measures

Read:

- Obsolescence is mildly negatively correlated with every measure of new innovation.
- The correlations among the new-innovation measures themselves are positive and often larger.
- This is one of the paper’s most important descriptive results because it shows that the measure is related to innovation but not redundant with the standard innovation proxies.

Table 3
Relation between Obsolescence and measures of new innovation

Main reference		Sample period	Firm sample	Patent sample			
Patent Value (\$M)	Kogan et al. (2017)	1926–2020	All U.S. public firms	All patents			
Citation-weighted patents (CW)	(KFS)			From U.S. public firms			
% Breakthrough patents	Kelly et al. (2021)	1840–2002	N/A	All patents			
RETech	Bowen, Fritsch, and Hoberg (2023)	1930–2017	VC-backed U.S. startups	All patents			
Tech Breadth	(BFT)						
Text-Based Innovation	Bellem, Bhagat, and Cookson (2021)	1989–2010	All S&P 500 firms	All patents from S&P 500 firms			
(BIC)							
B. Correlation Between Obsolescence and Innovation Measures							
	Obsolescence	Patent Value (\$M)	Citation-Weighted Patents (CW)	% Breakthrough Patents	RETech	Tech Breadth	Text-Based Innovation
Obsolescence	1	–0.061	–0.085	–0.148	–0.105	–0.005	–0.083
Patent Value (\$M)		1	0.307	0.157	0.265	–0.152	0.378
Citation-Weighted Patents (CW)			1	0.151	0.101	–0.009	0.22
% Breakthrough Patents				1	0.373	–0.21	–0.403
RETech					1	–0.413	0.227
Tech Breadth						1	–0.303
Text-Based Innovation							1

This table examines the correlation between Obsolescence and other innovation measures. Patent Number is the number of patents that the firm files in that year. Avg. Citation Per Patent is the average citations of those patents in that year. Patent Value (SM) is the stock market-based patent value from Kogan et al. (2017). Citation-Weighted Patents (CW) is the citation-weighted patent count following Kogan et al. (2017). % Breakthrough Patents is the share of breakthrough patents of firms’ patents in that year and the definition of breakthrough patents is from Kelly et al. (2021). RETech measures whether the patent pertains to a technological area that is rapidly evolving (i.e., following breakthroughs) or stable from Bowen, Fritsch, and Hoberg (2023). Tech Breadth measures how much (or little) the patent’s text is spread across technological fields from Bowen, Fritsch, and Hoberg (2023). Text-Based Innovation and Neg. Text-Based Innovation is the text-based innovation measure and negative text-based innovation measure from Bellem, Bhagat, and Cookson (2021). For each measure pair, we use the overlap firm-year observations to calculate the correlation.

Table 4
Technological obsolescence and firm growth

Time horizon =	$t+1$	$t+2$	$t+3$	$t+4$	$t+5$
Profits ($N=21,274$)					
$Obsolescence_t$	–0.013*** (0.003)	–0.021*** (0.005)	–0.025*** (0.007)	–0.029*** (0.009)	–0.029** (0.011)
R^2	0.251	0.269	0.268	0.260	0.252
Output ($N=22,358$)					
$Obsolescence_t$	–0.009** (0.004)	–0.017*** (0.006)	–0.022*** (0.008)	–0.027** (0.010)	–0.031** (0.013)
R^2	0.201	0.211	0.211	0.208	0.208
Capital ($N=22,873$)					
$Obsolescence_t$	–0.013*** (0.003)	–0.024*** (0.005)	–0.034*** (0.008)	–0.044*** (0.010)	–0.051*** (0.012)
R^2	0.178	0.196	0.202	0.202	0.198
Labor ($N=23,511$)					
$Obsolescence_t$	–0.006** (0.002)	–0.012** (0.004)	–0.016** (0.007)	–0.018** (0.009)	–0.018* (0.010)
R^2	0.170	0.179	0.182	0.181	0.181
TFP ($N=16,639$)					
$Obsolescence_t$	–0.007* (0.003)	–0.010** (0.005)	–0.013** (0.005)	–0.016*** (0.005)	–0.014** (0.006)
R^2	0.236	0.293	0.327	0.341	0.344

This table examines the relation between *Obsolescence* and firm growth and productivity using the model below (Equation (2)) in the paper:

$$\log Y_{f,t+\tau} - \log Y_{f,t} = \beta_1 \cdot Obsolescence_{f,t} + \theta_1 \cdot X_{f,t} + \delta_1 I_{X,t} + \varepsilon_{f,t+\tau}.$$

The outcome variables, Y , include firm profits, output, capital, employment, and TFP. All defined and described in Table 1. The table presents results estimated using up to 5 years from t . Controls include the level $\log Y_{f,t}$, the log value of the capital stock, the log number of employees, and the log number of patents granted up to year t , the log value of the firm age, and the firm’s idiosyncratic volatility. All right-hand-side variables are standardized to unit standard deviation to facilitate magnitude interpretations. The model includes industry (SIC3)-by-year fixed effects. Standard errors are clustered by firm and year, and they are shown in parentheses. *, **, and *** denote statistical significance at the 10%, 5%, and 1% levels, respectively.

5.8 Table 4: Baseline effect on firm growth and productivity

Read:

- One standard deviation higher obsolescence predicts lower future growth almost everywhere.
- At the five-year horizon, the coefficients imply about 2.9 percentage points lower profit growth, 3.1 percentage points lower output growth, 5.1 percentage points lower capital growth, 1.8 percentage points lower labor growth, and 1.4 percentage points lower TFP growth.
- Capital responds the most, which fits a creative-destruction view in which obsolete firms gradually lose scale and investment capacity.
- The coefficients get more negative with horizon, meaning the effects are persistent and cumulative rather than immediate and short-lived.

5.9 Table 5: Baseline results controlling for new innovation

Read:

- The obsolescence coefficients barely change after controlling for SM and CW.
- Patent value (SM) enters positively and often more strongly than obsolescence, which is what we expect for a measure of new successful innovation.
- Citation-weighted patents are weaker and often insignificant, echoing the paper’s argument that raw citation-weighted patent counts are a fragile proxy.
- This table is central for the paper’s claim that technological obsolescence is distinct from the arrival of new innovation.

Table 5
Technological obsolescence and growth, controlling for innovation measures

Time horizon =	$t+1$	$t+2$	$t+3$	$t+4$	$t+5$
Profits ($N=21,274$)					
<i>Obsolescence_t</i>	-0.012*** (0.003)	-0.019*** (0.005)	-0.022*** (0.007)	-0.025*** (0.009)	-0.025** (0.011)
<i>Citation-Weighted Patents_t</i> (CW)	0.002 (0.005)	0.005 (0.008)	0.009 (0.011)	0.007 (0.014)	0.011 (0.016)
<i>Patent Value_t</i> (SM)	0.017** (0.007)	0.026** (0.011)	0.035** (0.014)	0.045** (0.017)	0.055*** (0.018)
R^2	0.254	0.272	0.272	0.265	0.258
Output ($N=22,358$)					
<i>Obsolescence_t</i>	-0.008** (0.004)	-0.016** (0.006)	-0.021** (0.008)	-0.024** (0.010)	-0.028** (0.013)
<i>Citation-Weighted Patents_t</i> (CW)	-0.005 (0.004)	-0.005 (0.007)	-0.007 (0.010)	-0.008 (0.012)	-0.007 (0.015)
<i>Patent Value_t</i> (SM)	0.014** (0.006)	0.021* (0.011)	0.029** (0.014)	0.035* (0.019)	0.046** (0.020)
R^2	0.202	0.213	0.213	0.211	0.212
Capital ($N=22,873$)					
<i>Obsolescence_t</i>	-0.013*** (0.003)	-0.022*** (0.005)	-0.032*** (0.008)	-0.041*** (0.010)	-0.048*** (0.012)
<i>Citation-Weighted Patents_t</i> (CW)	-0.010*** (0.002)	-0.013*** (0.004)	-0.014** (0.006)	-0.013 (0.008)	-0.014 (0.010)
<i>Patent Value_t</i> (SM)	0.016*** (0.004)	0.028*** (0.008)	0.036*** (0.012)	0.042** (0.016)	0.049** (0.018)
R^2	0.185	0.203	0.208	0.207	0.204
Labor ($N=23,511$)					
<i>Obsolescence_t</i>	-0.006** (0.002)	-0.011** (0.005)	-0.015** (0.007)	-0.016* (0.009)	-0.017 (0.010)
<i>Citation-Weighted Patents_t</i> (CW)	-0.005* (0.003)	-0.007 (0.005)	-0.008 (0.007)	-0.010 (0.010)	-0.014 (0.011)
<i>Patent Value_t</i> (SM)	0.011*** (0.004)	0.018** (0.007)	0.023** (0.009)	0.028** (0.012)	0.033** (0.014)
R^2	0.173	0.182	0.185	0.184	0.184
TFP ($N=16,639$)					
<i>Obsolescence_t</i>	-0.006* (0.003)	-0.008* (0.004)	-0.011** (0.005)	-0.014*** (0.005)	-0.012* (0.006)
<i>Citation-Weighted Patents_t</i> (CW)	0.000 (0.003)	0.006 (0.006)	0.008 (0.007)	0.005 (0.008)	0.006 (0.009)
<i>Patent Value_t</i> (SM)	0.015* (0.009)	0.019 (0.012)	0.024* (0.012)	0.030** (0.011)	0.033*** (0.010)
R^2	0.239	0.296	0.331	0.346	0.350

This table examines the relation between *Obsolescence* and firm growth and productivity after adding measures of new innovation, the stock market-based patent value from Kogan et al. (2017), and citation-weighted patent counts. The design follows that in Table 4. *, **, and *** denote statistical significance at the 10%, 5%, and 1% levels, respectively.

Table 6
Technological obsolescence and growth, isolating variations in *Obsolescence*

Time horizon =	$t+1$	$t+2$	$t+3$	$t+4$	$t+5$
Profits ($N=21,274$)					
<i>Obsolescence_t</i>	-0.016*** (0.003)	-0.025*** (0.005)	-0.029*** (0.007)	-0.032*** (0.008)	-0.033*** (0.009)
<i>Citation-Weighted Patents_t</i> (CW)	0.001 (0.005)	0.003 (0.008)	0.006 (0.011)	0.005 (0.014)	0.008 (0.016)
<i>Patent Value_t</i> (SM)	0.015** (0.007)	0.024** (0.011)	0.033** (0.013)	0.043** (0.016)	0.052*** (0.018)
R^2	0.255	0.273	0.273	0.266	0.259
Output ($N=22,358$)					
<i>Obsolescence_t</i>	-0.010*** (0.004)	-0.021*** (0.005)	-0.025*** (0.007)	-0.030*** (0.010)	-0.037*** (0.012)
<i>Citation-Weighted Patents_t</i> (CW)	-0.006 (0.004)	-0.007 (0.007)	-0.009 (0.010)	-0.010 (0.012)	-0.010 (0.015)
<i>Patent Value_t</i> (SM)	0.013** (0.005)	0.020* (0.010)	0.027* (0.013)	0.033* (0.019)	0.043** (0.019)
R^2	0.203	0.213	0.214	0.211	0.213
Capital ($N=22,873$)					
<i>Obsolescence_t</i>	-0.016*** (0.002)	-0.028*** (0.004)	-0.041*** (0.007)	-0.053*** (0.009)	-0.063*** (0.011)
<i>Citation-Weighted Patents_t</i> (CW)	-0.011*** (0.002)	-0.014*** (0.004)	-0.017** (0.006)	-0.017* (0.008)	-0.018* (0.010)
<i>Patent Value_t</i> (SM)	0.015*** (0.004)	0.026*** (0.008)	0.033*** (0.011)	0.037** (0.015)	0.044** (0.017)
R^2	0.187	0.205	0.211	0.211	0.208
Labor ($N=23,511$)					
<i>Obsolescence_t</i>	-0.008*** (0.003)	-0.014*** (0.004)	-0.020*** (0.006)	-0.022*** (0.007)	-0.027*** (0.009)
<i>Citation-Weighted Patents_t</i> (CW)	-0.006* (0.003)	-0.008 (0.005)	-0.010 (0.008)	-0.011 (0.010)	-0.016 (0.011)
<i>Patent Value_t</i> (SM)	0.010*** (0.003)	0.017** (0.007)	0.021** (0.009)	0.026** (0.012)	0.031** (0.014)
R^2	0.173	0.182	0.185	0.185	0.185
TFP ($N=16,639$)					
<i>Obsolescence_t</i>	-0.008** (0.004)	-0.012** (0.005)	-0.012** (0.006)	-0.013** (0.006)	-0.011 (0.007)
<i>Citation-Weighted Patents_t</i> (CW)	-0.000 (0.003)	0.005 (0.006)	0.008 (0.007)	0.004 (0.008)	0.005 (0.010)
<i>Patent Value_t</i> (SM)	0.014 (0.008)	0.018 (0.012)	0.023** (0.011)	0.029** (0.011)	0.032*** (0.010)
R^2	0.239	0.296	0.331	0.346	0.350

This table examines the relation between *Obsolescence* and firm growth and productivity using a new construction. Specifically, this approach reconstructs the *Obsolescence* measure with one modification: instead of using all patents in a firm’s technology base when tracking changes in citations between $t - \tau$ and t , we use only patents in the base that experienced a decline of citations over this time window. The table design follows that in Table 5. *, **, and *** denote statistical significance at the 10%, 5%, and 1% levels, respectively.

5.10 Table 6: Isolating the true obsolescence part of the measure

Read:

- The author reconstructs the measure using only patents in the technological base that experienced citation declines.
- The negative coefficients remain, and in several cases they become even larger in magnitude than in Table 5.
- This greatly strengthens the interpretation that the real-side results are driven by actual technological deterioration rather than by unrelated noise in citation dynamics.

5.11 Table 7: Heterogeneity across patent type and product-market competition

Read:

- Obsolescence in **core** technologies matters much more than obsolescence in peripheral technologies.
- Obsolescence in **product** / **embodied** innovation is generally more damaging than obsolescence in process / disembodied innovation.
- The effects are stronger in industries with higher product-market competition.
- This table is economically important because it shows the paper is not just finding a reduced-form negative correlation; the pattern lines up with clear economic mechanisms.

Table 7
Heterogeneity across different firm and industry characteristics

Heterogeneity	Core Patents		Product/Process Patents		Competition	
	Core	Noncore	Product	Process	High	Low
Profits						
$Obsolescence_t$	-0.021*** (0.007)	-0.006 (0.007)	-0.025*** (0.007)	-0.010* (0.005)	-0.024*** (0.008)	-0.026* (0.015)
R^2	0.267	0.266	0.268	0.267	0.242	0.383
Observations	21,274	21,274	21,274	21,274	15,513	5,761
Output						
$Obsolescence_t$	-0.022*** (0.008)	-0.005 (0.007)	-0.022** (0.008)	-0.008 (0.005)	-0.023** (0.010)	-0.018 (0.012)
R^2	0.211	0.210	0.211	0.210	0.189	0.348
Observations	22,358	22,358	22,358	22,358	16,564	5,794
Capital						
$Obsolescence_t$	-0.030*** (0.007)	-0.014** (0.006)	-0.034*** (0.008)	-0.011* (0.006)	-0.038*** (0.009)	-0.014 (0.010)
R^2	0.202	0.199	0.202	0.198	0.184	0.297
Observations	22,873	22,873	22,873	22,873	17,022	5,851
Labor						
$Obsolescence_t$	-0.013** (0.006)	-0.006 (0.006)	-0.018** (0.007)	-0.006 (0.005)	-0.017** (0.008)	-0.007 (0.010)
R^2	0.182	0.181	0.182	0.181	0.156	0.307
Observations	23,511	23,511	23,511	23,511	17,525	5,986
TFP						
$Obsolescence_t$	-0.013*** (0.004)	-0.007 (0.004)	-0.013** (0.005)	-0.008** (0.004)	-0.014** (0.006)	-0.007 (0.008)
R^2	0.327	0.327	0.327	0.327	0.302	0.474
Observations	16,639	16,639	16,639	16,639	12,804	3,835

This table examines the relation between *Obsolescence* and firm growth and productivity in different subsamples. The core versus noncore (peripheral) patents are defined as the top technology class(es) that populate 50% of all the firm's patents. The product (disembodied) vs. process (embodied) innovation is defined using the textual description of patents based on [Bena and Simintzi \(2025\)](#). The product market competition is categorized into high versus low based on the SIC3 HHI. The empirical design follows that in [Table 4](#), only the $t+3$ horizon is reported. *, **, and *** denote statistical significance at the 10%, 5%, and 1% levels, respectively.

Table 8
Technological obsolescence and competitor innovation measures

Time horizon =	$t+1$	$t+2$	$t+3$	$t+4$	$t+5$
Profits ($N = 21,274$)					
$Obsolescence_t$	-0.012*** (0.003)	-0.020*** (0.005)	-0.023*** (0.007)	-0.026*** (0.009)	-0.026** (0.011)
$Patent\ value_t$ (SM)	0.017** (0.007)	0.027** (0.011)	0.036** (0.013)	0.045*** (0.016)	0.055*** (0.018)
$Competitors' patent\ value_t$	-0.001 (0.005)	-0.011 (0.007)	-0.013** (0.006)	-0.025*** (0.008)	-0.024** (0.009)
R^2	0.254	0.272	0.272	0.266	0.258
Output ($N = 22,358$)					
$Obsolescence_t$	-0.007** (0.004)	-0.015** (0.006)	-0.020*** (0.008)	-0.024** (0.010)	-0.027** (0.013)
$Patent\ value_t$ (SM)	0.013** (0.005)	0.020** (0.010)	0.028** (0.013)	0.034* (0.018)	0.045** (0.019)
$Competitors' patent\ value_t$	-0.003 (0.005)	-0.001 (0.008)	0.002 (0.008)	-0.005 (0.010)	-0.008 (0.010)
R^2	0.202	0.213	0.213	0.211	0.212
Capital ($N = 22,873$)					
$Obsolescence_t$	-0.012*** (0.003)	-0.021*** (0.005)	-0.031*** (0.008)	-0.040*** (0.010)	-0.047*** (0.013)
$Patent\ value_t$ (SM)	0.014*** (0.004)	0.025*** (0.007)	0.033*** (0.011)	0.038** (0.014)	0.046*** (0.016)
$Competitors' patent\ value_t$	-0.010** (0.004)	-0.011 (0.008)	-0.015 (0.009)	-0.019* (0.011)	-0.019 (0.012)
R^2	0.183	0.202	0.208	0.207	0.204
Labor ($N = 23,511$)					
$Obsolescence_t$	-0.006** (0.002)	-0.010** (0.005)	-0.014* (0.007)	-0.015* (0.009)	-0.016 (0.010)
$Patent\ value_t$ (SM)	0.010*** (0.003)	0.016** (0.006)	0.021** (0.008)	0.026** (0.011)	0.030** (0.013)
$Competitors' patent\ value_t$	0.000 (0.004)	-0.000 (0.007)	-0.001 (0.008)	-0.002 (0.008)	-0.004 (0.009)
R^2	0.172	0.181	0.184	0.184	0.183
TFP ($N = 16,639$)					
$Obsolescence_t$	-0.006* (0.003)	-0.008* (0.004)	-0.012** (0.005)	-0.014*** (0.005)	-0.012* (0.006)
$Patent\ value_t$ (SM)	0.014 (0.009)	0.019 (0.013)	0.024** (0.012)	0.031*** (0.011)	0.033*** (0.010)
$Competitors' patent\ value_t$	-0.011 (0.008)	-0.017** (0.007)	-0.000 (0.009)	0.004 (0.008)	-0.004 (0.007)
R^2	0.239	0.296	0.331	0.346	0.350

This table examines the relation between *Obsolescence* and firm growth and productivity after adding competitors' innovation value (the stock market-based patent value from KPSS), which is defined as the value of patents created by firms in the same SIC3 industry except the focal firm itself. The design follows that in [Table 4](#). *, **, and *** denote statistical significance at the 10%, 5%, and 1% levels, respectively.

5.12 Table 8: Comparison with competitor innovation measures

Read:

- After controlling for the firm's own patent value and for competitors' patent value, the obsolescence coefficient remains large and statistically significant.
- Competitor innovation does help predict weaker outcomes in some cases, but the paper's obsolescence measure is generally more robust.
- Economically, this is exactly what we should expect if technological destruction often comes from outside the narrow set of same-industry public competitors.

5.13 Table 9: Monthly returns of obsolescence-sorted portfolios

Read:

- Raw value-weighted excess returns decline monotonically from the low-obsolescence portfolio to the high-obsolescence portfolio.
- The low-minus-high spread earns about 30 basis points per month in raw excess return.
- The factor-adjusted alphas are much larger: depending on the model, the low-minus-high strategy earns roughly 36 to 59 basis points per month. In the five-factor-plus-momentum style specification, the alpha is around 57 basis points per month, or about 7% annually.
- After correcting for the fact that the paper’s innovative-firm sample itself has a positive alpha, the results remain large. For example, with the sample-specific market factor, the four-factor alpha of the low-minus-high spread is about 43.6 basis points per month.
- Panel C shows that the low-obsolescence portfolio looks like a growth portfolio and the high-obsolescence portfolio looks more value-like; the spread loads negatively on HML.

Table 9
Monthly returns of obsolescence-sorted portfolios

A. Value-weight portfolio, with factor models							
	Event	3F	4F	4F + RMW + CMA	Q5 (DEZ)	3F ^{INT}	4F ^{INT}
Low	0.889*** (0.268)	0.388*** (0.085)	0.384*** (0.084)	0.413*** (0.086)	0.336*** (0.095)	0.423*** (0.095)	0.410*** (0.094)
Middle	0.639*** (0.239)	0.065 (0.066)	0.106 (0.067)	0.007 (0.067)	-0.100 (0.060)	0.078 (0.073)	0.028 (0.068)
High	0.587*** (0.235)	-0.029 (0.103)	0.027 (0.103)	-0.157 (0.096)	-0.194* (0.112)	-0.082 (0.105)	-0.144 (0.097)
Low-High	0.302 (0.178)	0.418*** (0.152)	0.377*** (0.150)	0.570*** (0.148)	0.539*** (0.161)	0.505*** (0.159)	0.588*** (0.150)
All	0.706*** (0.238)	0.148*** (0.047)	0.173*** (0.048)	0.101*** (0.045)	0.087 (0.055)	0.134*** (0.052)	0.154*** (0.055)
B. Value-weight portfolio, with factor models and sample-specific market factor							
	Event	3F	4F	4F + RMW + CMA	Q5 (DEZ)	3F ^{INT}	4F ^{INT}
Low	0.889*** (0.268)	0.294*** (0.087)	0.270*** (0.084)	0.336*** (0.086)	0.330*** (0.095)	0.316*** (0.091)	0.284*** (0.089)
Middle	0.639*** (0.239)	-0.105*** (0.040)	-0.089** (0.039)	-0.090** (0.039)	-0.108** (0.046)	-0.110** (0.041)	-0.092** (0.039)
High	0.587*** (0.235)	-0.190** (0.080)	-0.166** (0.080)	-0.240*** (0.078)	-0.201** (0.083)	-0.211** (0.082)	-0.240*** (0.077)
Low-High	0.302 (0.178)	0.497*** (0.149)	0.436*** (0.146)	0.770*** (0.145)	0.731*** (0.159)	0.728*** (0.157)	0.828*** (0.155)

(Continued)

Table 9
(Continued)

Value-weight portfolio four-factor loadings				
	MKT	SM	HML	UMD
Low	0.971*** (0.022)	-0.053 (0.052)	-0.344*** (0.030)	0.005 (0.025)
Middle	0.869*** (0.029)	-0.132*** (0.023)	-0.068** (0.027)	-0.055** (0.027)
High	0.842*** (0.030)	-0.031 (0.032)	0.153** (0.068)	-0.017 (0.040)
Low-High	0.130 (0.061)	-0.022 (0.064)	-0.497*** (0.087)	0.079 (0.058)
All	0.963*** (0.020)	-0.093*** (0.056)	-0.100*** (0.021)	-0.013* (0.018)

This table presents monthly portfolio returns (in %) for portfolios sorted on *Obsolescence*. At the end of June of year t from 1986 to 2016, we sort firms based on their obsolescence measure into three portfolios (Low, Middle, and High). The Low portfolio contains all stocks below the 30th percentile in *Obsolescence*, and the High portfolio contains all stocks above the 70th percentile. We also construct a portfolio containing all the stocks in our sample, labeled as All. The *Obsolescence* used to form these portfolios are from the prior calendar year $t-1$. Based on our formation of the technological obsolescence, the measure is publicly observable at the end of year $t-1$ and does not incorporate any forward-looking information. We hold these portfolios over the next 12 months, from July of year t to June of year $t+1$. We compute their value-weighted monthly returns. We report the average monthly return in excess of 1-month Treasury-Bill rate (Event). We also report alphas from the regression of the time series of portfolio excess returns on various factor models: the Fama-French three factors (3F), the four factors (4F, three factors + UMD/momentum), 4F + RMW + CMA (value-minus-risk, conservative-minus-aggressive), the q -Factor model (Hui, Xie, and Zhang 2015), and the factor models that replace the traditional Fama-French HML factor with the intangible-adjusted HML (Easkin, Kim, and Papell 2022). In panel A we report the results for value-weight portfolios. In panel B we include a sample-specific market factor in the regressions to correct for the sample-specific abnormal return. In panel C we report the four-factor loadings of the portfolios. Standard errors are reported in parentheses.

5.14 Table 10: Technological obsolescence and analyst forecast errors

Read:

- Forecast errors are significantly more negative for high-obsolescence firms.
- With new-innovation controls, the obsolescence coefficient is about -0.469 for one-year-ahead EPS forecast errors and -0.771 for two-year-ahead forecast errors.
- The sign is exactly what the underreaction story predicts: analysts are too optimistic about the future profits of firms whose technologies are becoming obsolete.
- This table is the key bridge between the firm’s real decline and the stock-return anomaly.

Table 10
Technological obsolescence and forecasting errors

	(1) $(\pi_{f,t+1} - F_t \pi_{f,t+1}) / P_{f,t-1}$	(2)	(3) $(\pi_{f,t+2} - F_t \pi_{f,t+2}) / P_{f,t-1}$	(4)
Obsolescence	-0.514** (0.216)	-0.469** (0.212)	-0.793** (0.339)	-0.771** (0.347)
Observations	22,846	22,846	20,624	20,624
R ²	0.024	0.024	0.032	0.032
New innovation control	No	Yes	No	Yes

This table reports the results from regressing firm-level EPS forecast errors on *Obsolescence* based on Equation (3). The dependent variables are the forecast errors based on the consensus 1-year and 2-year forecasts for the current fiscal year earnings, that is, $(\pi_{f,t+\tau} - F_t \pi_{f,t+\tau}) / P_{f,t-1}$ for $\tau = 1, 2$. In all columns, we control the log value of the capital stock, the log number of employees, and the log number of patents granted up to year t , the log value of the firm age, and the firm’s idiosyncratic volatility. Columns 2 and 4 have additional controls of new innovation measures, including the stock market-based patent value from Kogan et al. (2017), and citation-weighted patent counts. Standard errors are clustered by firm and year, and they are shown in parentheses. *, **, and *** denote statistical significance at the 10%, 5%, and 1% levels, respectively.

6 Short summary

- **Conclusion.** The paper builds a new measure of technological obsolescence from citation declines to a firm’s technological base and shows that it predicts both weaker firm performance and lower stock returns.
- **Intuition.** When the technologies a firm has been standing on stop being useful for later innovation, the firm loses growth opportunities, productivity, and eventually investor support — but markets do not fully realize that immediately.
- **Empirical lesson.** Innovation is not just about the arrival of new ideas. The depreciation and destruction of old ideas are separately measurable and economically important.

7 Conclusion

7.1 Core conclusion (what the paper establishes)

- **Measurement contribution.** The paper provides a tractable, patent-based, firm-year measure of the destruction side of innovation.
- **Real-side result.** Firms with more obsolescent technologies grow more slowly in profits, output, capital, labor, and productivity.
- **Cross-sectional pattern.** The effects are strongest when obsolescence hits economically central technologies — core patents and product / embodied innovations — and in industries where product-market competition is strong.
- **Financial-markets result.** High-obsolescence firms underperform low-obsolescence firms by large abnormal-return margins.
- **Mechanism result.** Analyst forecast errors show that investors appear to underreact to technological obsolescence.

7.2 Economic intuition

- **Knowledge begets knowledge.** If the technologies a firm relied on in the past are losing relevance, the firm stands on a weaker knowledge base going forward.
- **Technology adjustment is costly.** Firms cannot frictionlessly replace obsolete knowledge with new knowledge, new human capital, and new physical capital overnight.
- **Slow-moving deterioration is hard to price.** Because technological obsolescence is subtle and gradual, markets appear to process it imperfectly.

7.3 Takeaway

This is a measurement paper with meaningful substantive payoffs. Its main message is that the destruction side of innovation can be measured directly, that this destruction has large real consequences for firm dynamics, and that financial markets do not fully price it right away. The paper also opens a broader agenda: once technological obsolescence is measurable, researchers can study it at the level of firms, technologies, inventors, industries, regions, and asset prices.